

Cell Images in Drug Discovery



Anne Carpenter



Runxi Shen

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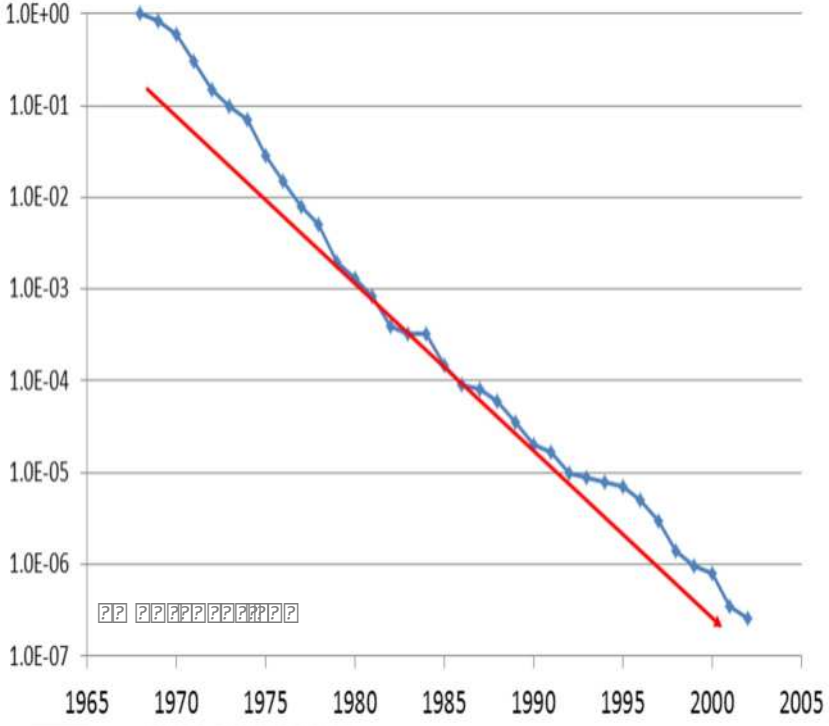
Download PDF of slides here: http://broad.io/CarpenterSlides

(talk content is non-confidential)

Tale of two industries

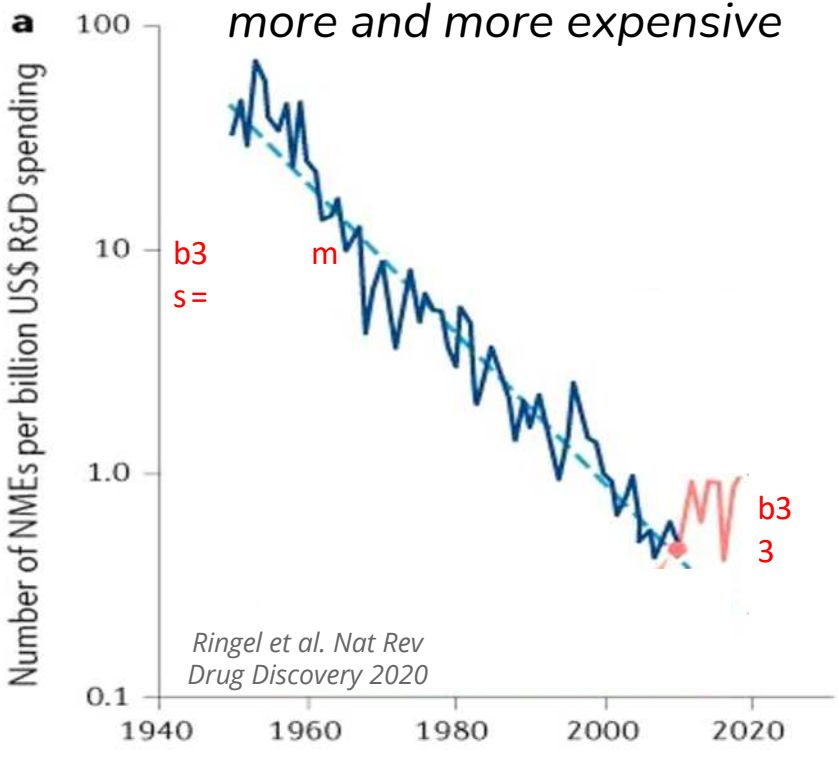
Moore's Law

Compute gets cheaper and cheaper

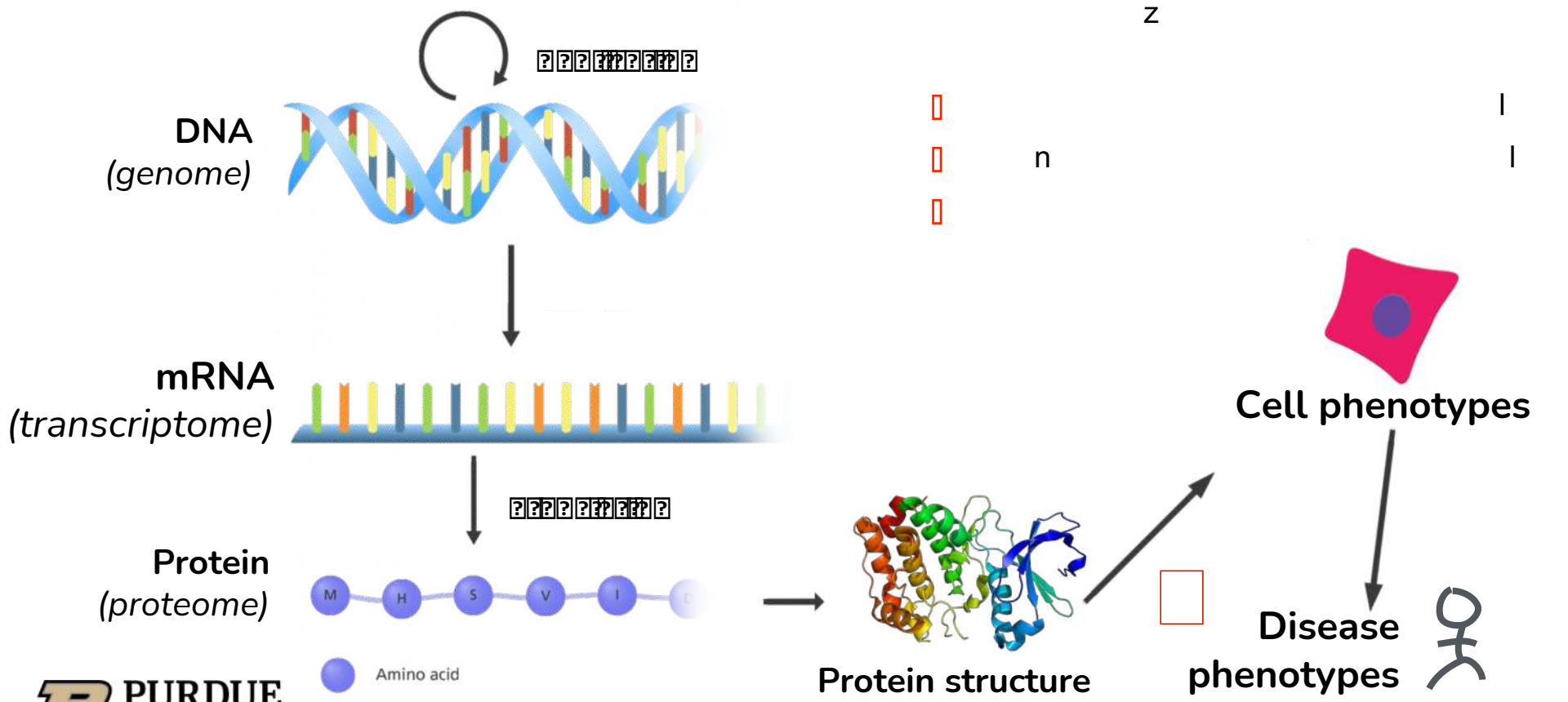


Eroom's Law

Discovering new medicines gets more and more expensive

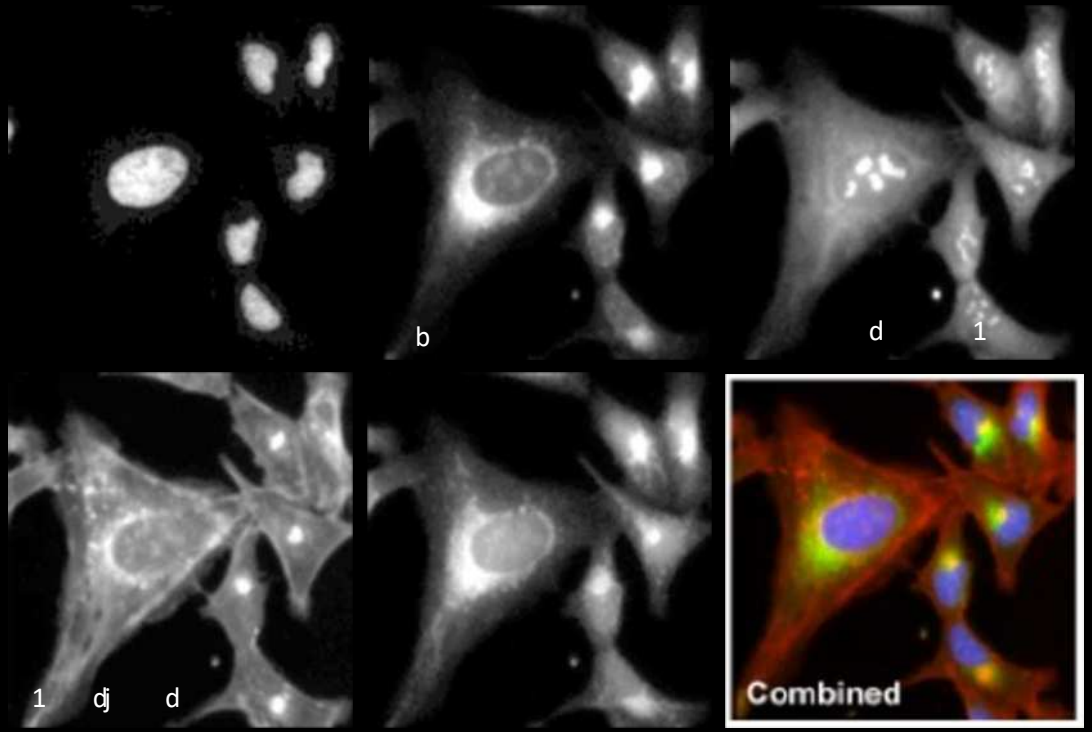


AI is getting better at some tasks faster than others



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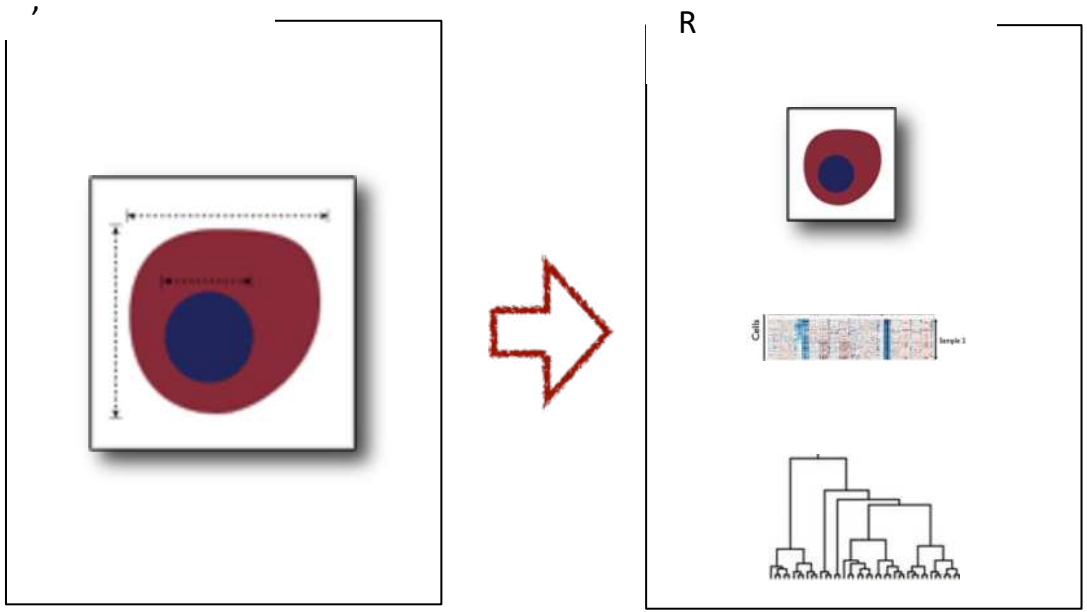
Images contain a wealth of information



1. single-cell resolution
2. complex model systems
3. multiplexed
4. quantitative

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A revolution in how we use images



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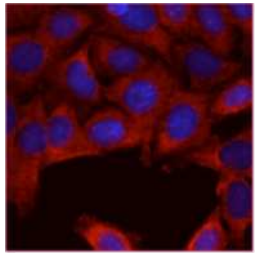
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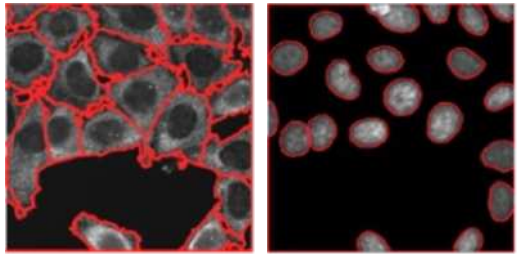
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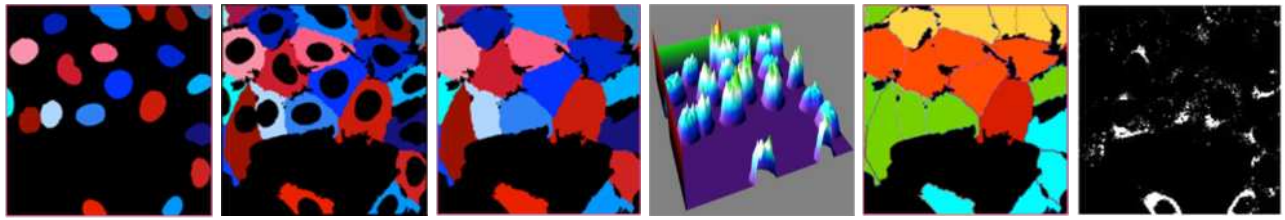
Extracting features from images







CellProfiler™
cell image analysis software



CellProfiler: Open-source image analysis software

CellProfiler™
cell image analysis software

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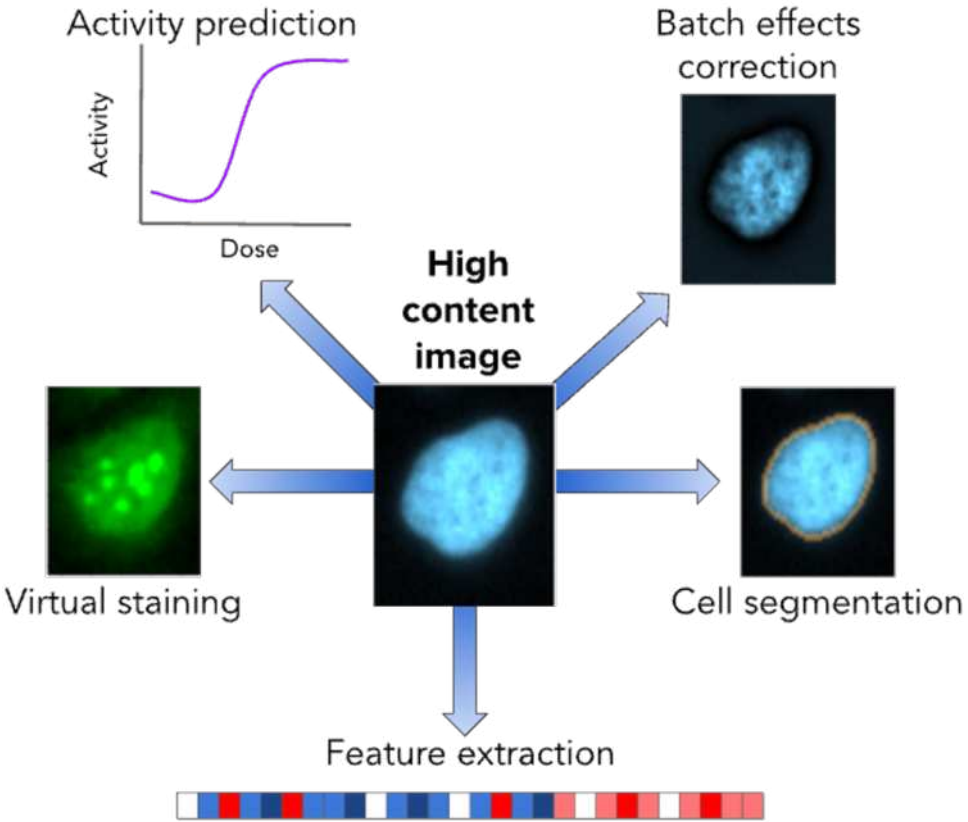

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image.sc

Impact of machine learning on image analysis



CellProfiler™
cell image analysis software

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OPIXIMI™

OH

image.sc

PC /



Beth Cimini's lab,
Broad Institute

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b b RT B P 5 frfM
P D4 p p D5 w b frfg

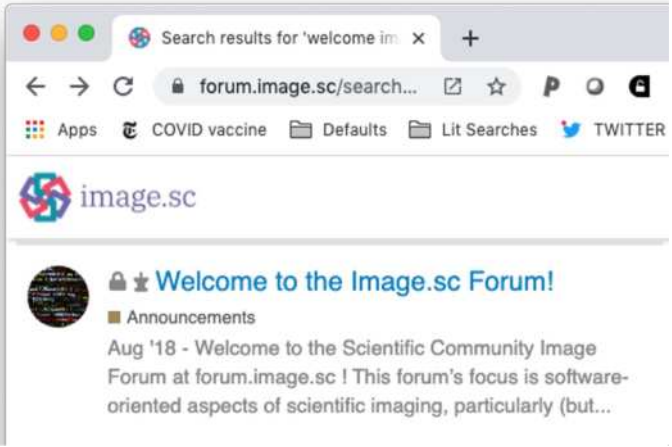
Just getting started in bioimage analysis?

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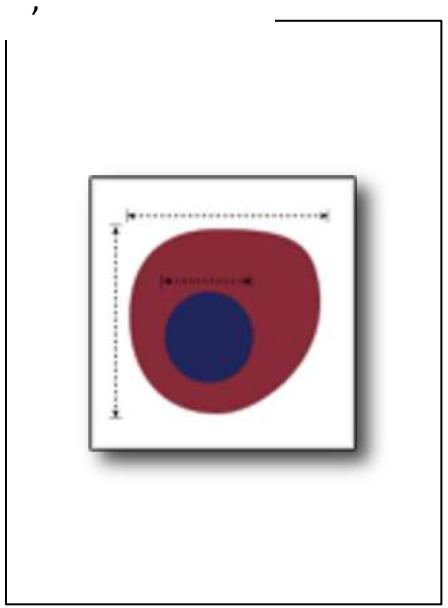


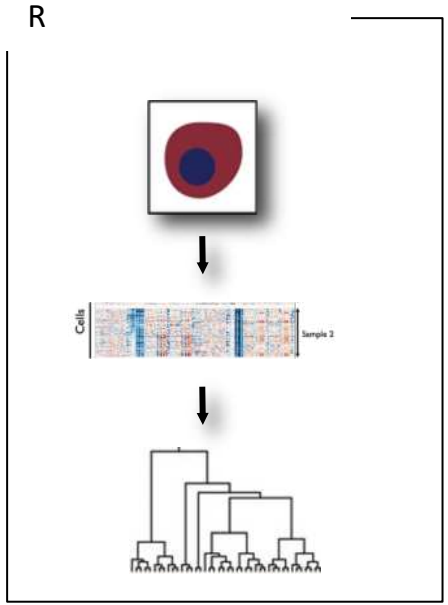


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A revolution in how we use images

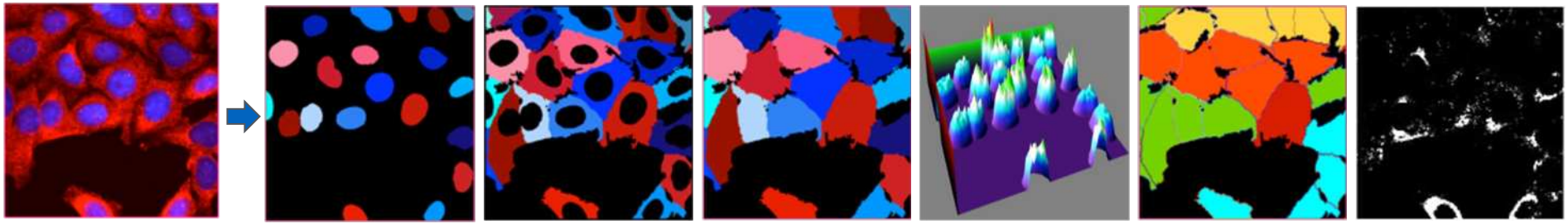




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Measure everything



CellProfiler™
cell image analysis software

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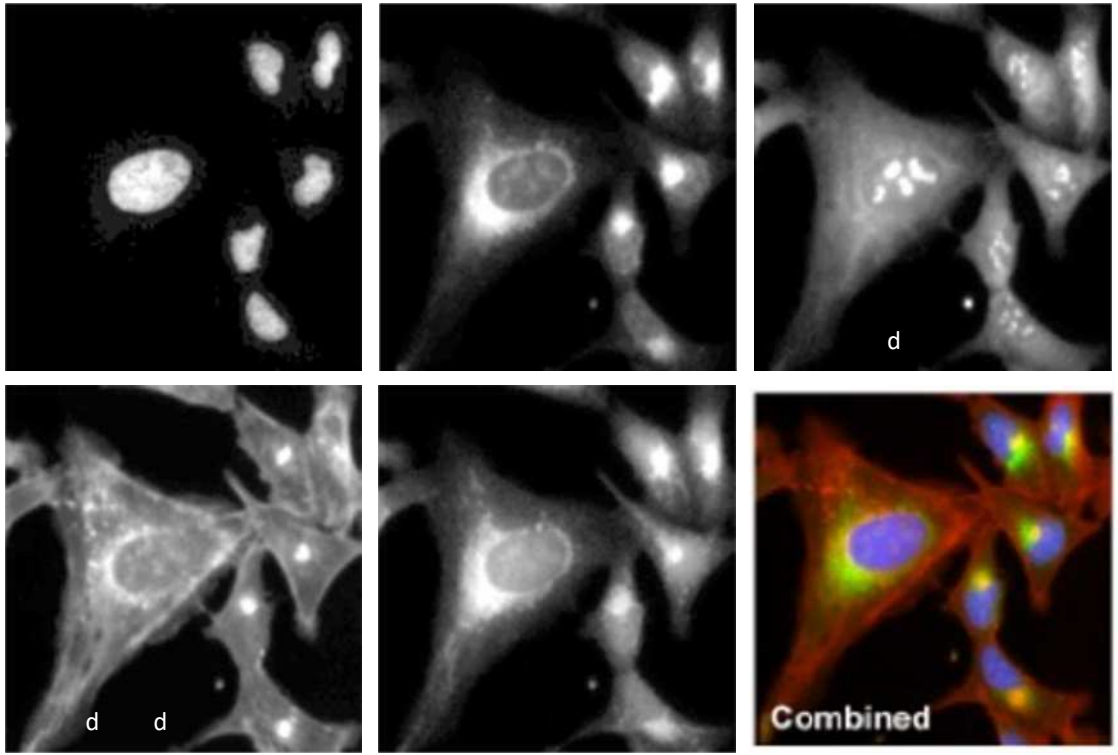
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The Cell Painting assay maximizes information for image-based profiling



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g l CPOS




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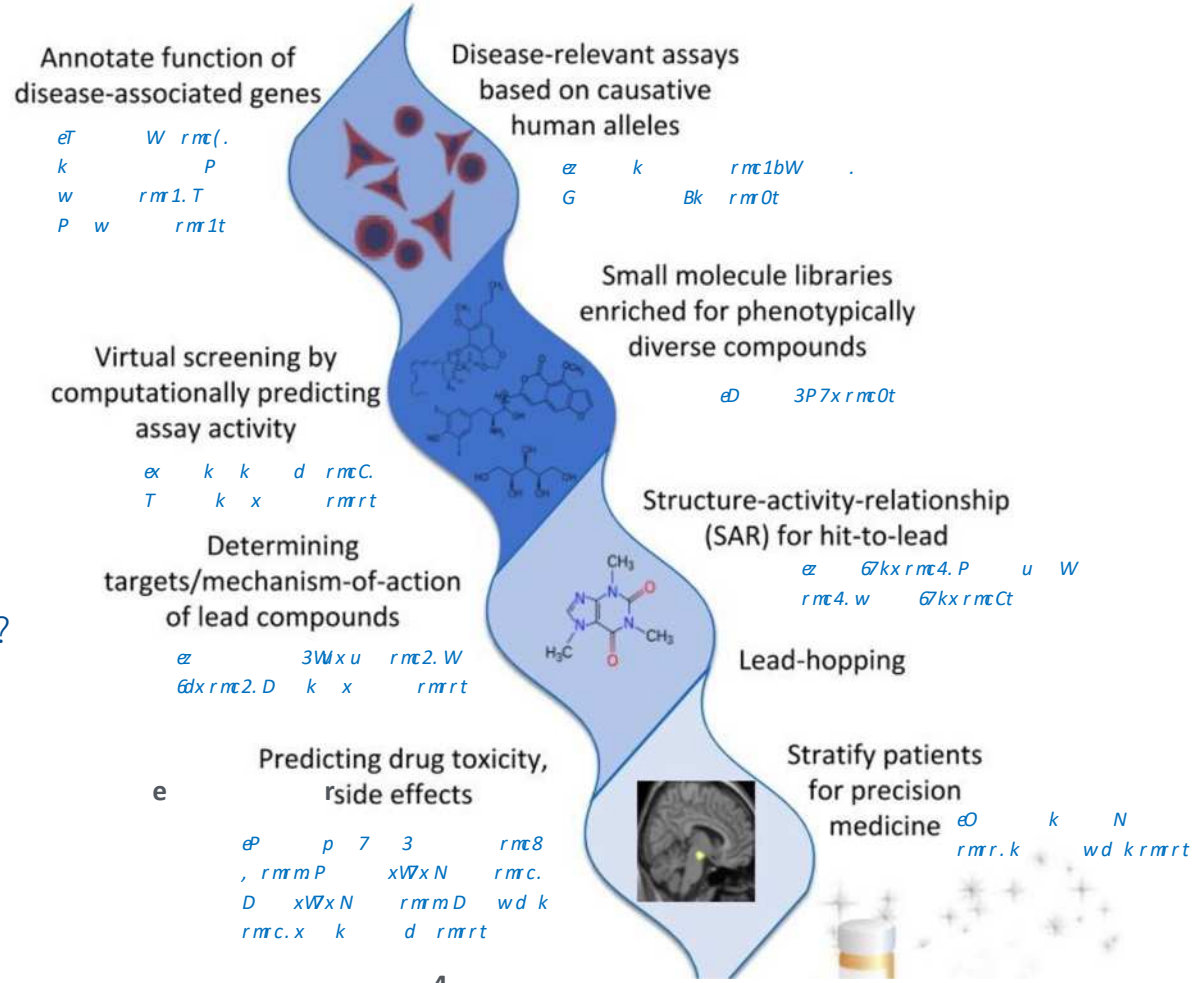

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Image-based profiling: accelerating many steps of drug discovery

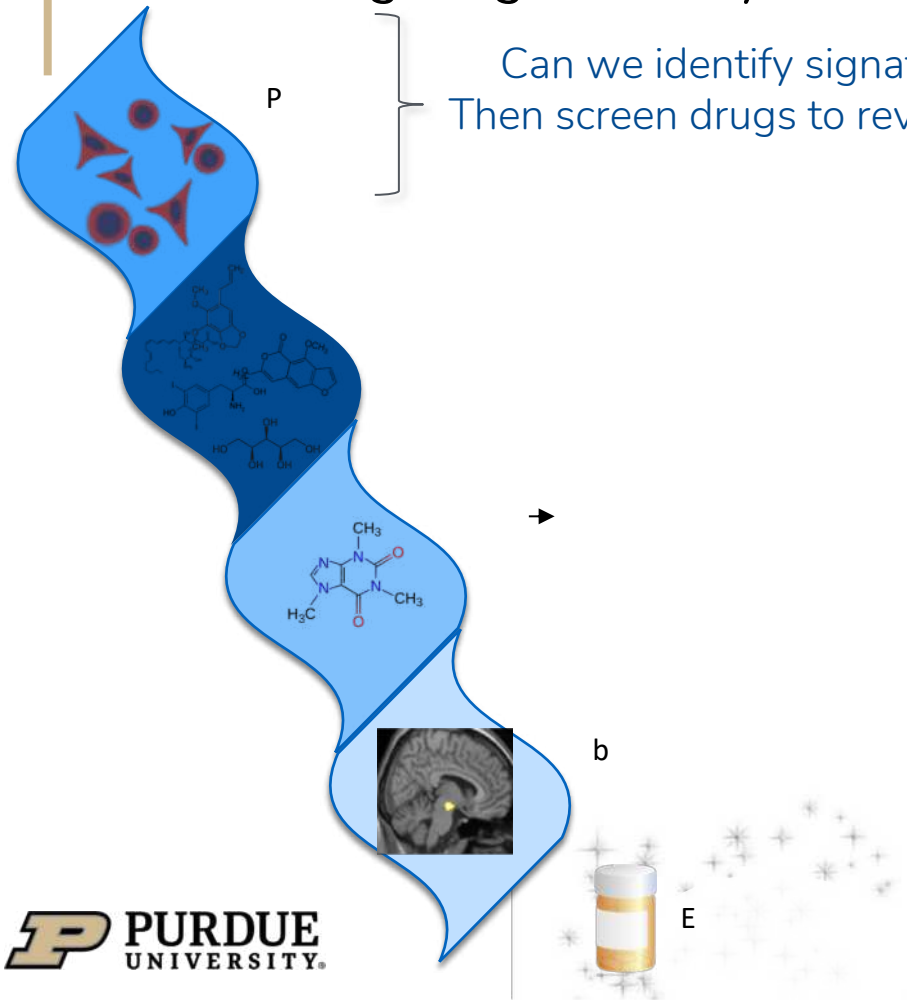
Can we speed up individual steps using automation, clever biotech, AI?

Can we **surpass human capabilities**?



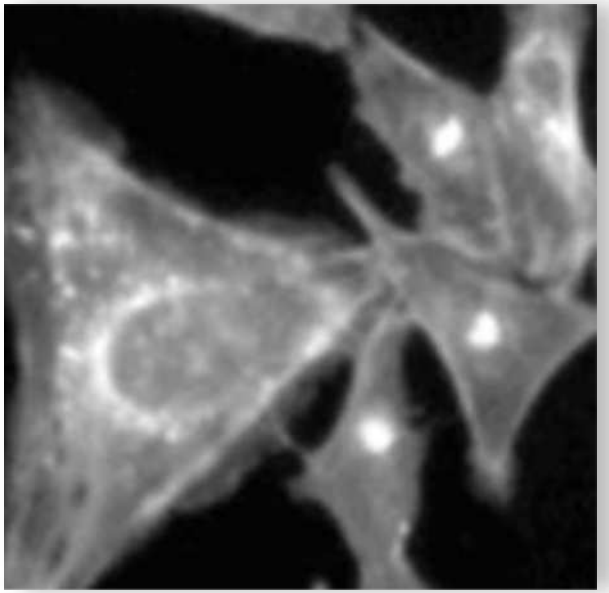
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Transforming drug discovery via image-based profiling



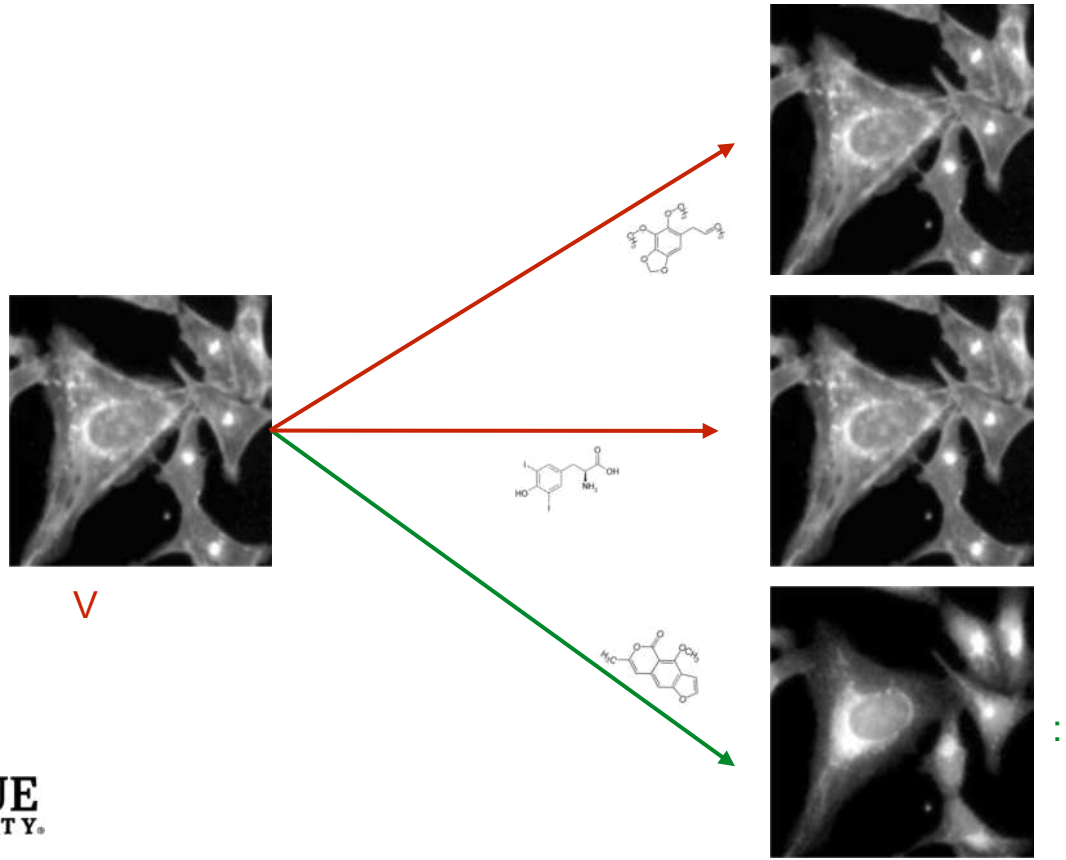
Can we identify signatures of disease?
Then screen drugs to reverse the signature?

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If cell images can tell us they're diseased… then we could screen drugs at scale



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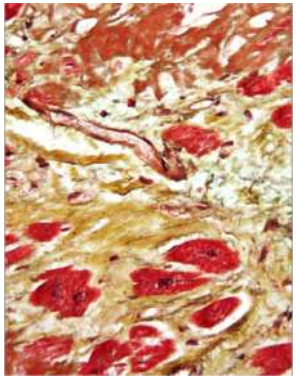


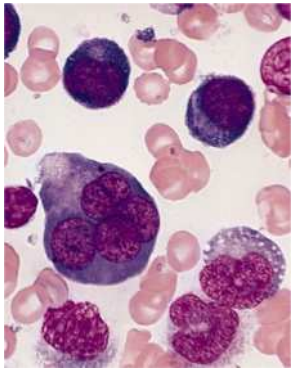
Prediction:Down's
Confidence:0.999

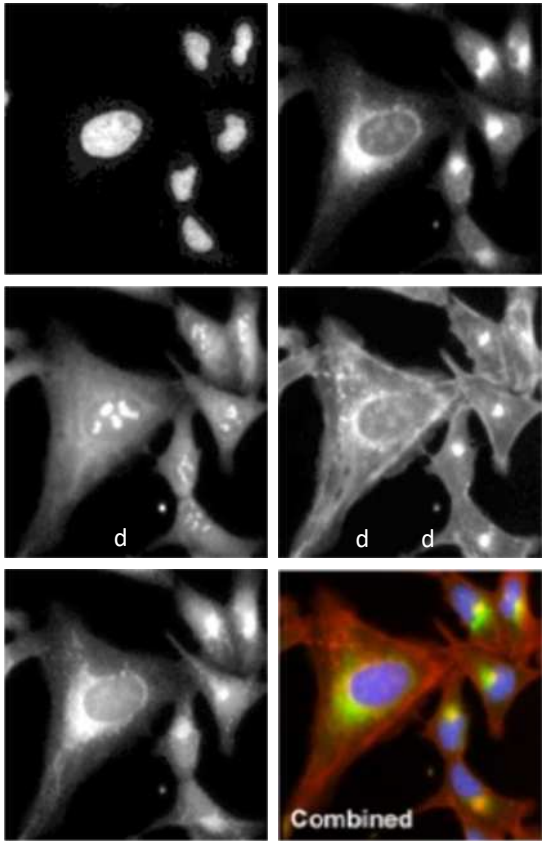


Prediction:Angelman
Confidence:0.997





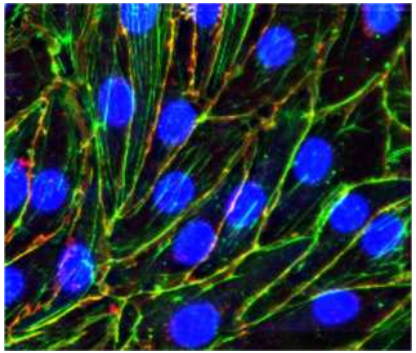




The Cell Painting assay

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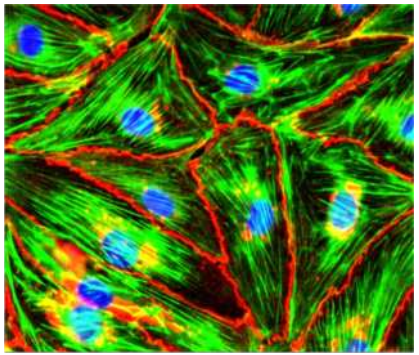
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Can we find drugs that reverse every disease phenotype?

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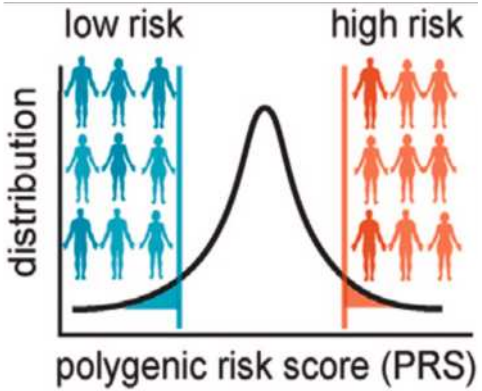
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Obesity example: patient cells at high vs. low risk
We can identify disease phenotypes using patient cell lines



LipocyteProfiler: Melina Claussnitzer lab

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Psychiatric example: patient cells with/without a condition







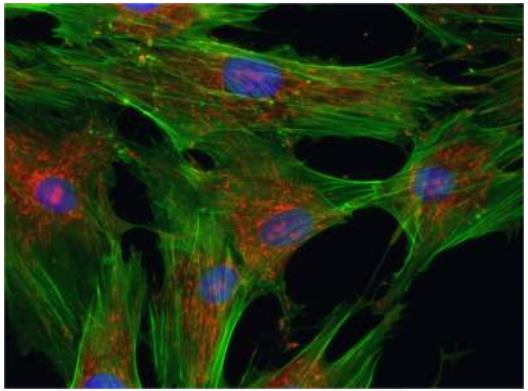
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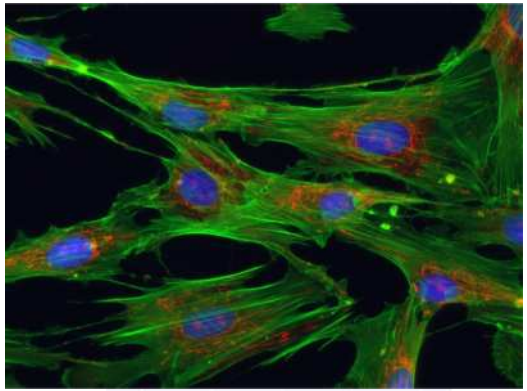
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Rare diseases present a major unmet need

RARE DISEASES AFFECT

30 MILLION

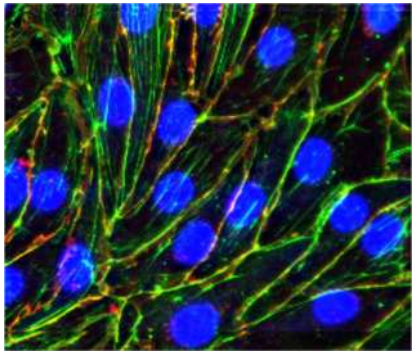
AMERICANS



APPROVED TREATMENTS ARE AVAILABLE FOR ONLY **5%** OF ALL RARE DISEASES

x

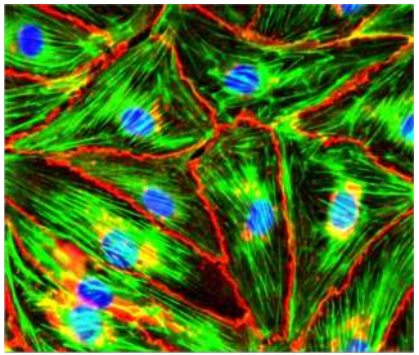
(more cheaply)



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Can we find drugs that reverse every disease phenotype?

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iGVF

“ H ’



Marc Vidal



Mike Calderwood



Anne Carpenter



Dave Hill



Fritz Roth



Mikko Taipale

Supported by a cast of dozens from these labs!

Current lead in Carpenter–Shen lab: Runxi Shen



BROAD INSTITUTE

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Sinai Health | Lunenfeld-Tanenbaum Research Institute

Donnelly Centre Cellular & Biomolecular Research UNIVERSITY OF TORONTO

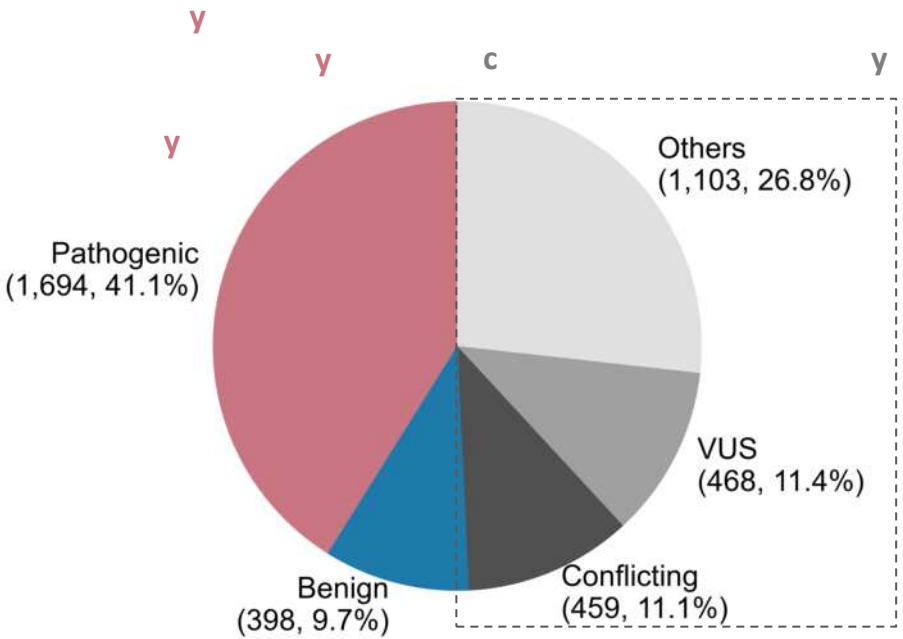
BLAVATNIK INSTITUTE HARVARD MEDICAL SCHOOL

CCSB Center for Cancer Systems Biology

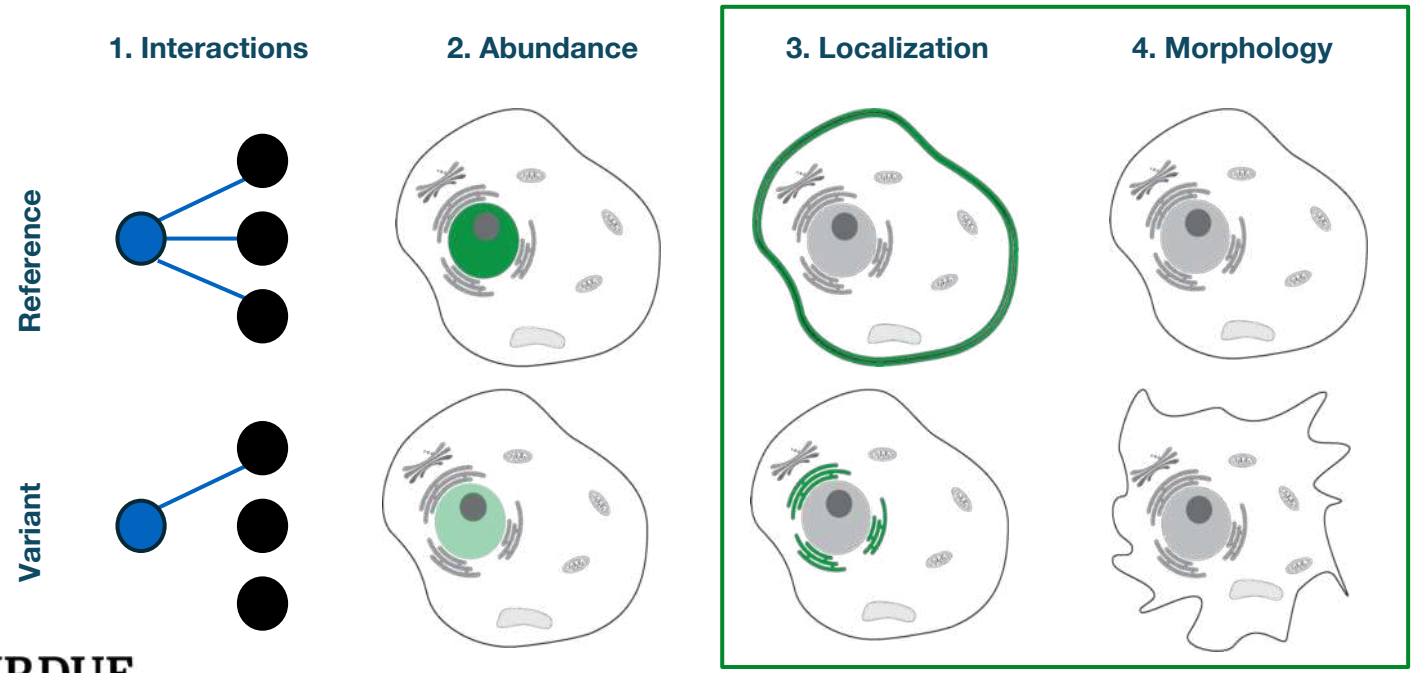
Dana-Farber Cancer Institute

VarChAMP aims to predict the impact of variants

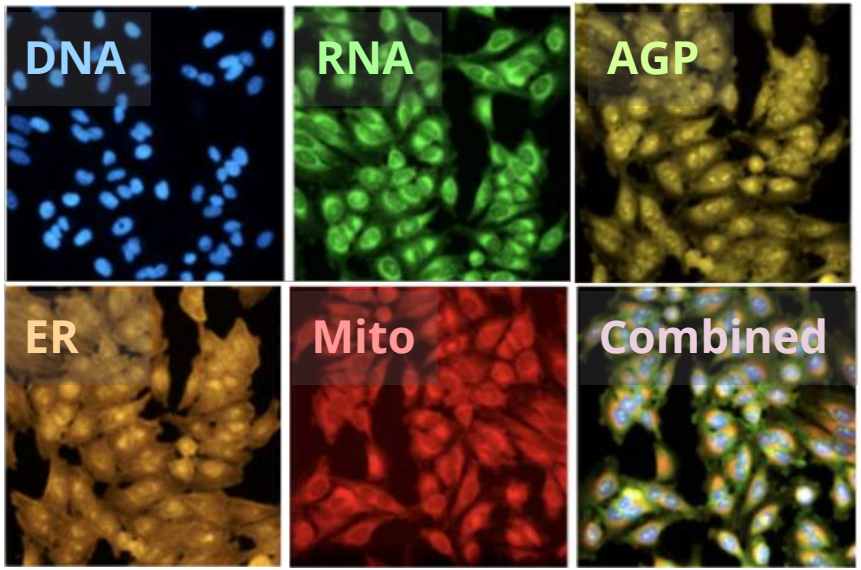
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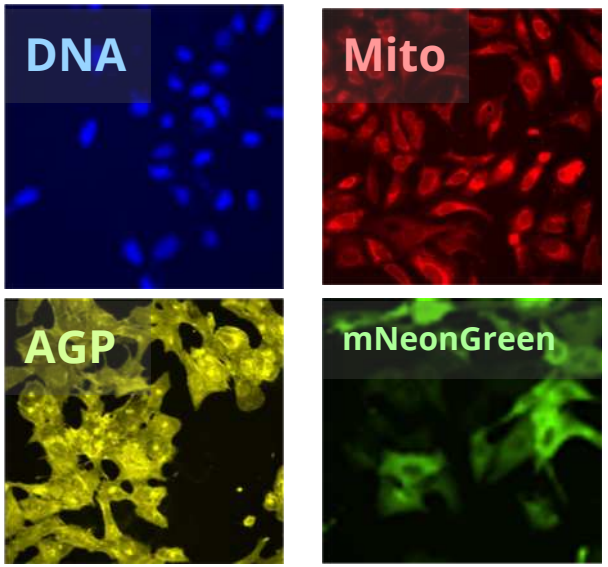
Capture the impact of coding variants at scale



Variant Painting Assay can profile genetic variants



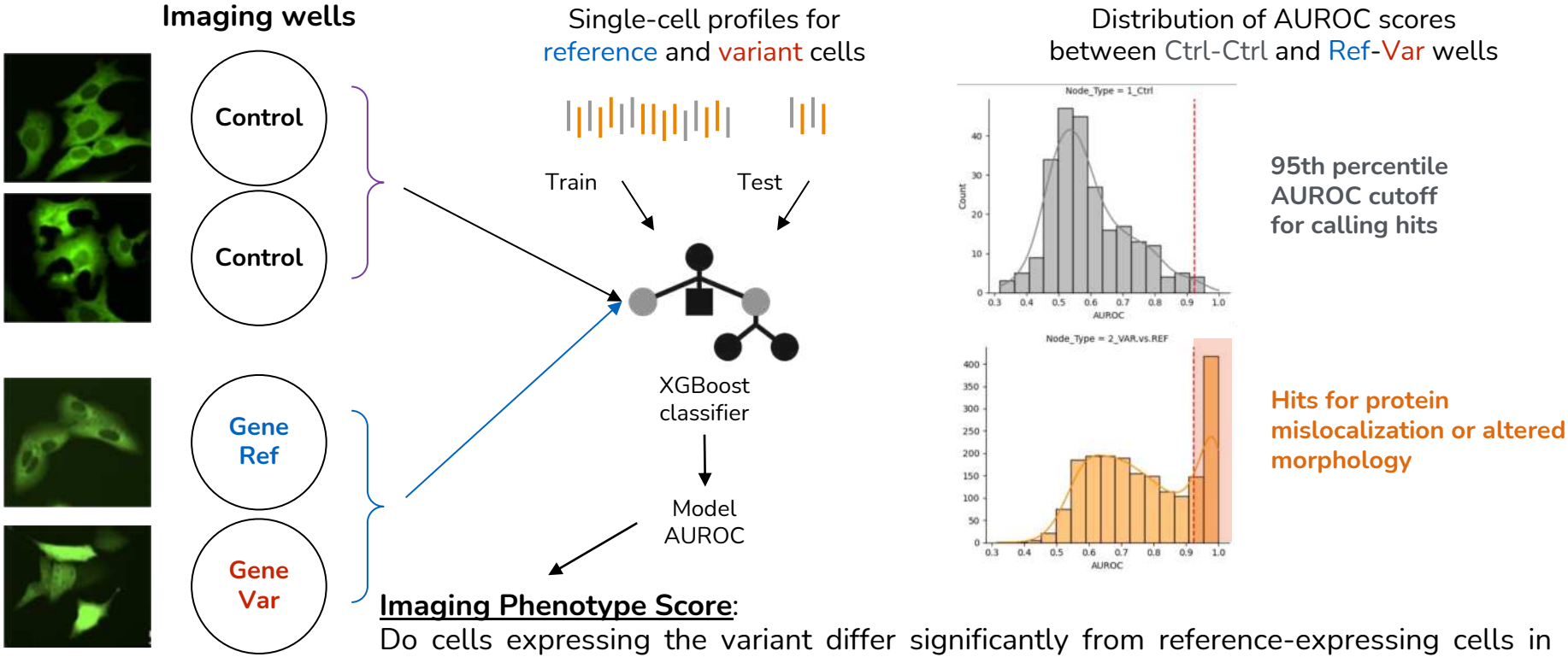
Standard Cell Painting
(general cell structures)



Variant Painting
(Morphology + Localization)

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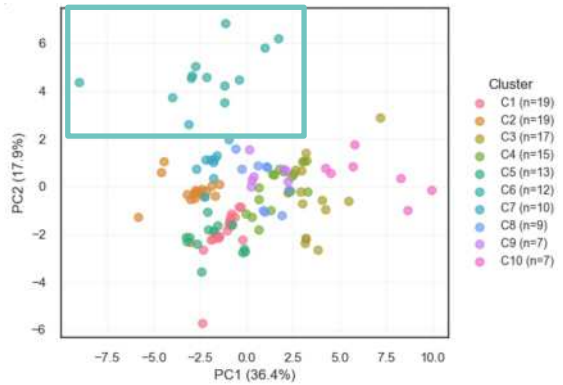
A machine learning workflow detects variant cells with imaging phenotypes



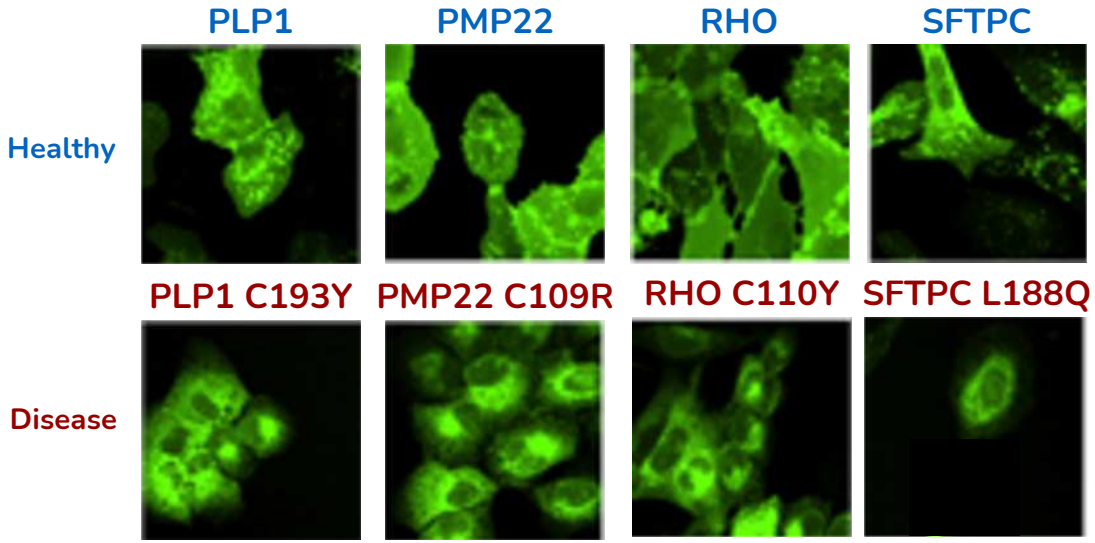
Imaging Phenotype Score:
Do cells expressing the variant differ significantly from reference-expressing cells in protein localization or morphology?

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A shared protein mislocalization phenotype could suggest a unified therapeutic strategy



Shared protein mislocalization patterns were observed across different genes, cell types, and disease modules.



Could all these shared mislocalization patterns potentially be rescued with the same drug?

- ❑ **TMED9-targeted clearance (Dvela-Levitt et al., Cell 2019)**
- ❑ **A compound to release the misfolded proteins trapped in TMED9 cargo receptor-enriched compartments**

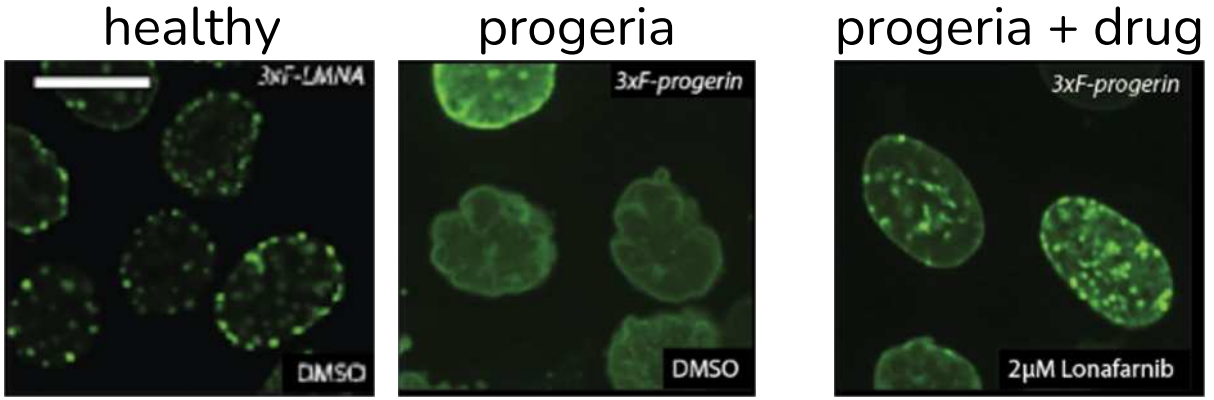
4,500 human genetic variants tested → 350+ disease phenotypes discovered

NIH IGVF Consortium: testing 30,000 new variants

VISTA Consortium: screen ~300 drugs x ~100 diseases

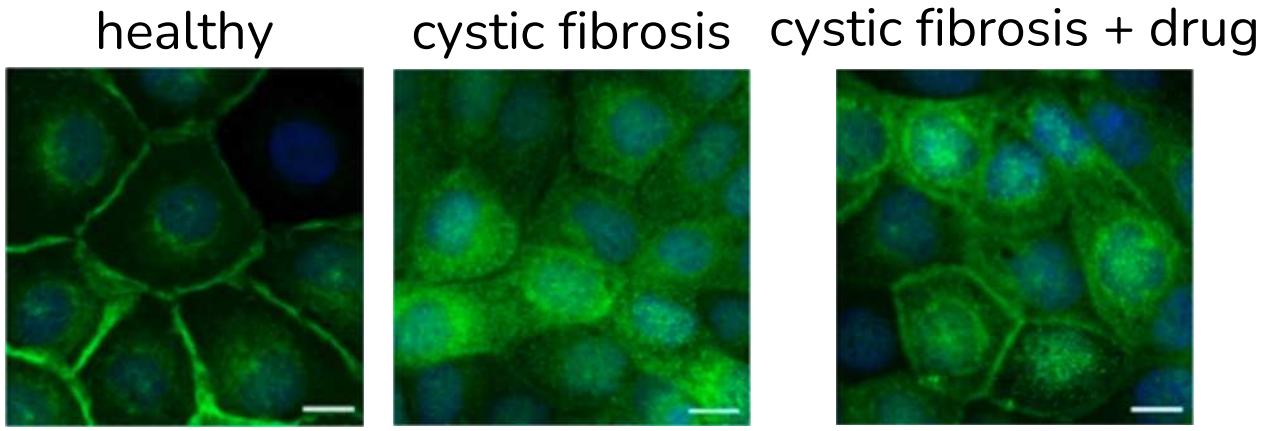
Some successful drugs reverse disease mislocalizations

Lonafarnib relocalizes LMNA in progeria syndrome (HGPS)



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Lumacaftor relocalizes CFTR in cystic fibrosis



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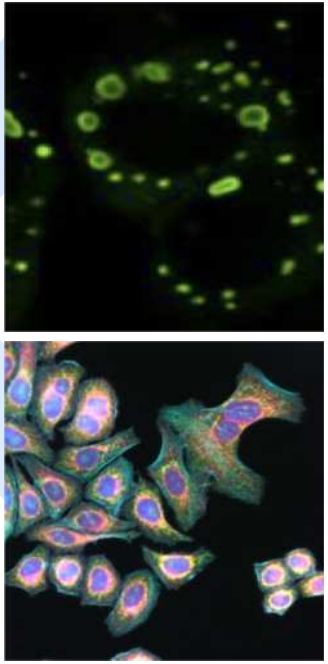
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Scaling up drug discovery:
Optical pooled screening (via in situ sequencing)



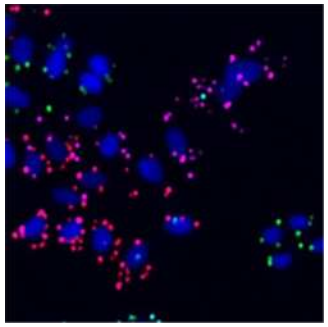
Each well =
1 compound +
200 reference genes
200 disease variants

Localization & morphology



+

in situ barcoding





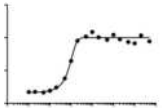
BF Ig OPL3

Workflow for finding drug-disease hypotheses…

1) Primary screen (pooled)



2) Confirmation screen for hits (non-pooled, dose response, replicates)



3) Confirm in disease-relevant cell lines



4) Engage the rare disease community of patients, clinicians, and researchers





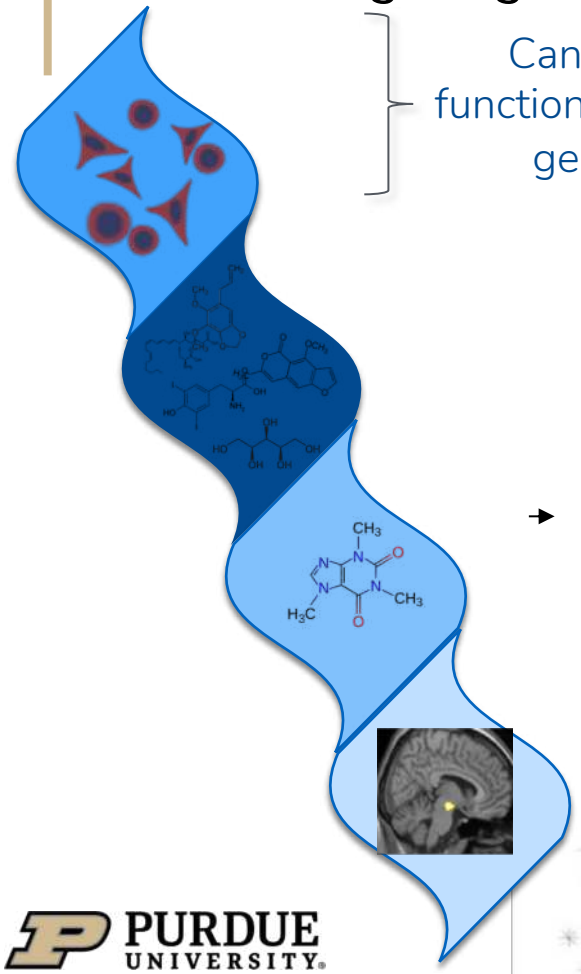


Cell-based experiments (disease-relevant cell types or assay readouts)
Animal experiments
Human clinical trials/off-label use
Encourage pharma investment for diseases with shared mislocalization

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Transforming drug discovery via Cell Painting

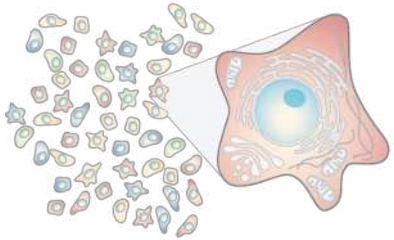
Can we identify gene functions by grouping similar genes and alleles?





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JUMP-Cell Painting Consortium dataset is now available



Partners:
- Broad Institute
- Amgen
- AstraZeneca
- Bayer
- Biogen
- Eisai
- Janssen
- Ksilink
- Merck Healthcare KGaA (Germany)
- Pfizer
- Servier
- Takeda

Supporting Partners:
- Nomic bio
- PerkinElmer
- Verily
- Google Research
- Horizon Discovery
- Ardigen

Collaborators:
- Pistoia Alliance
- Umea University
- Stanford University

JUMP Cell Painting data and tools now online!



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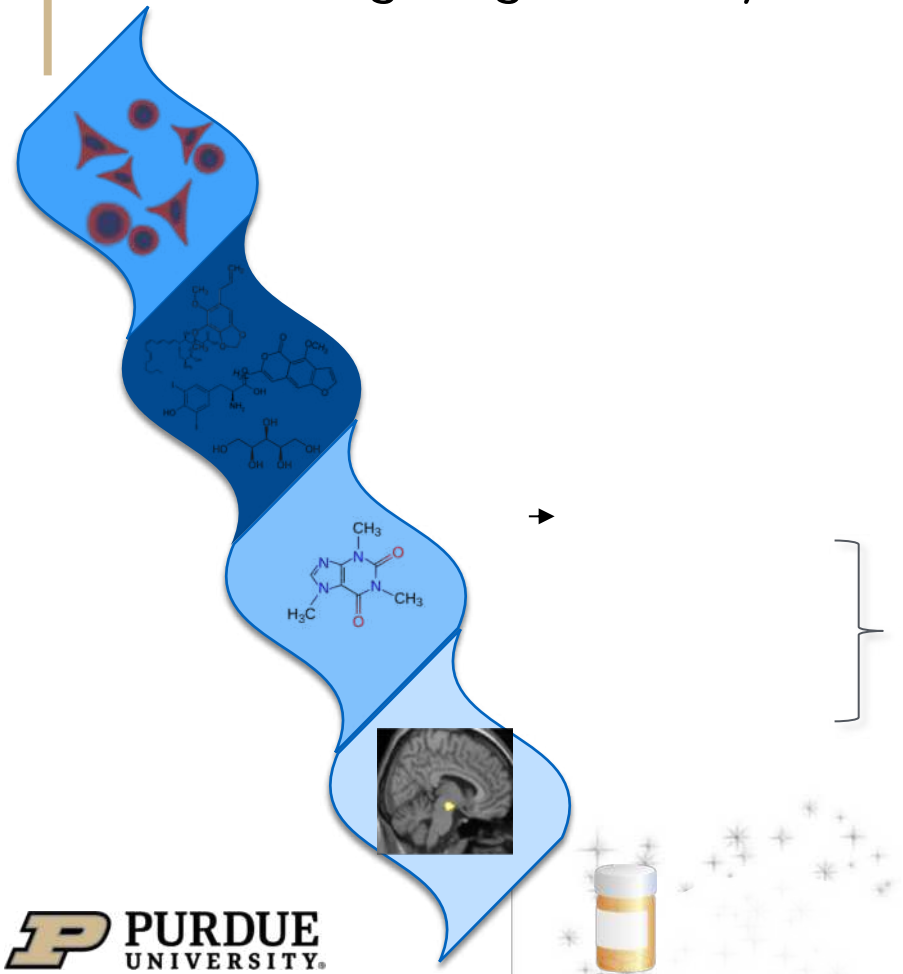
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Transforming drug discovery via Cell Painting



Can we identify various types of toxicity?

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Can we predict chemical toxicity from images?

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FDA NEWS RELEASE

FDA Announces Plan to Phase Out Animal Testing Requirement for Monoclonal Antibodies and Other Drugs

For Immediate Release: April 10, 2025

THE NIH DIRECTOR

The NIH Director

Statements

Testimonies

Advisory Groups

Previous NIH Directors

February 1, 2024

Statement on catalyzing the development of novel alternative methods

NIH National Institutes of Health
Office of Strategic Coordination–The Common Fund

Search Common Fund

Our Programs | Research Funding | News & Media | New Program Planning | Evaluation & Assessment

Complement Animal Research In Experimentation (Complement-ARIE) Program

OASIS Consortium for toxicity assessment

OASIS
Omics for Assessing Signatures for Integrated Safety Consortium

Public and private compounds with associated in vivo and/or human data

- 200 donated from industry partners
- 1400 from public domain

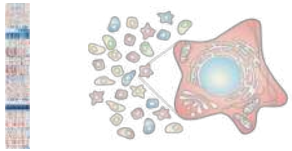
Cell model systems

(of increasing complexity, physiological relevance, and cost)

- U2OS human osteosarcoma immortal cell line
- HepaRG human hepatocyte-like cell line
- Primary rat hepatocytes
- Primary human hepatocytes
- Liver organoids, Liver-on-a-chip

Multi-omic profiling assays

Cell Painting (images)



Transcriptomics (mRNA)



Alithea, BioSpyder

nELISA (protein)

Nomic bio



Data analysis

- Translating *in vitro* to *in vivo* concentrations (qIVIVE)
- Distinguishing & interpreting mechanisms
- Integrating evidence across multiple data types/cell models

Publicly accessible data

Published methodologies

- Community of Practice



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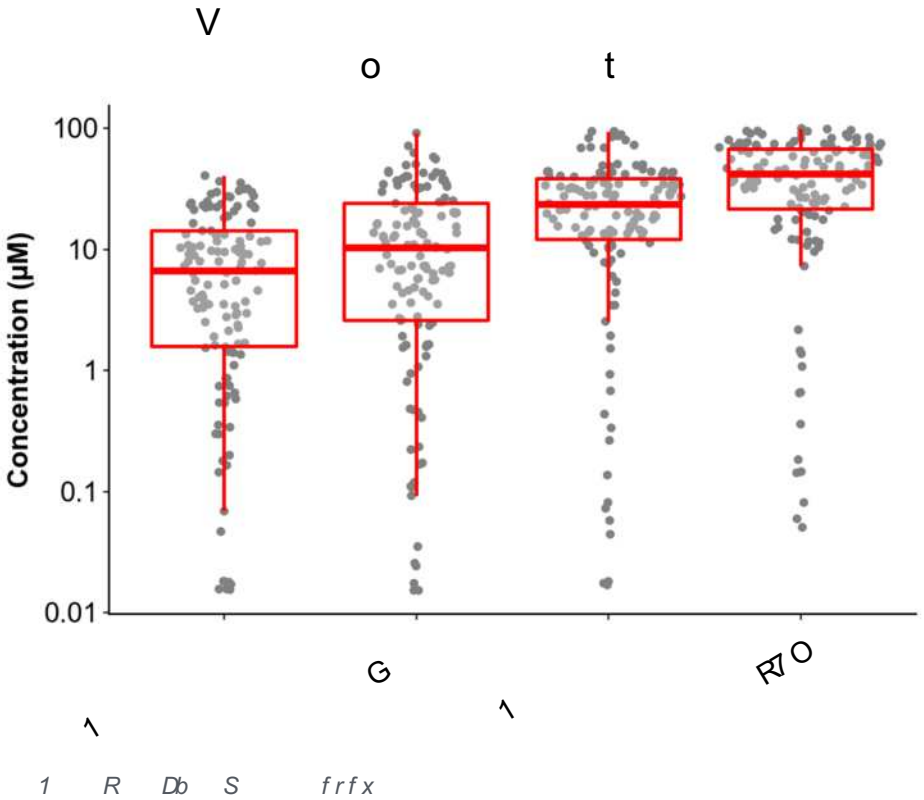
T R DE S frfy

Cell Painting detects bioactivity for more compounds and at lower concentrations than cytotoxicity assays



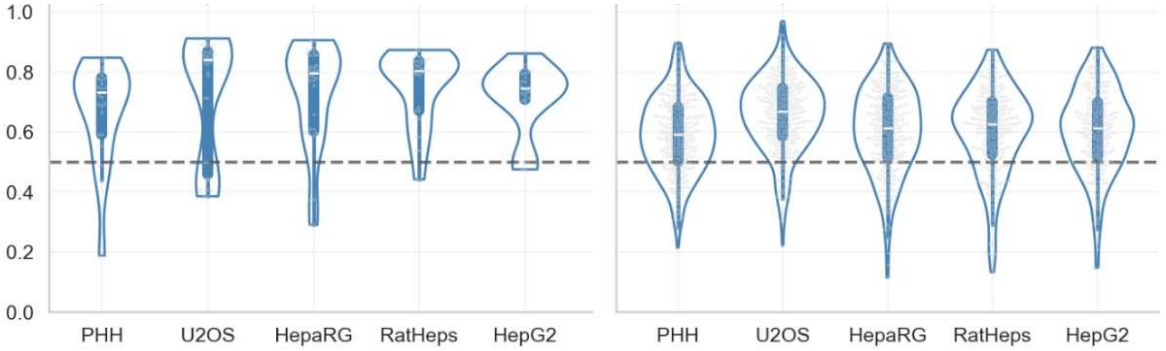
- Primary human hepatocytes
- 1085 compounds (8 doses, 2 replicates)
- Three assay readouts:
  - Biochemical assays (MTT and LDH)
  - Cell Painting assay (including cell count)

1 m1 b	**0t 7** m)5x b
G	**dnv** mnt x b
1	**ppt** m(nt x b
R7 O	**vd7** mgf x b



Morphology can predict cytotoxicity and drug modes of action across cell lines

OO	5t 6ngt %) m)5x b
(	56%ngf) r m6x b
O U	6r t ngf) r m)t x b
O U(	gf) m r 5t mt x b
O	r) 5r5(%m)6x b



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O

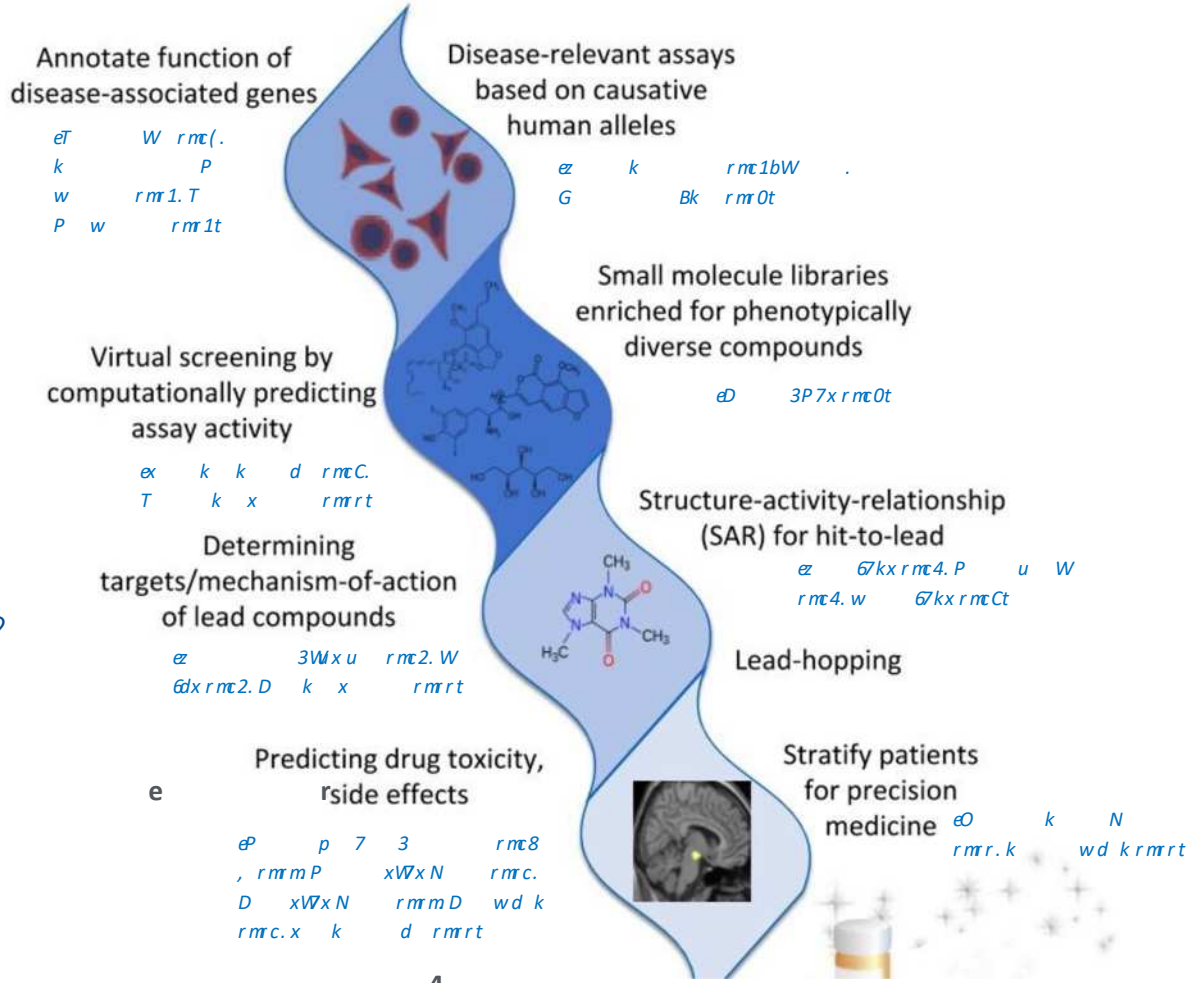
Can we speed up individual steps
using automation, clever biotech, AI?

Can we **surpass human capabilities**?

; uM r
vH 4 4 cCdCdrg
r vH P cCdCkn

S u M r v
H P cCd: Nn

; uM 4 r SIC



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